TRANSACTION TRACING & FRAUD DETECTION

Group - 12

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INTRODUCTION

In today’s digital world, large volumes of online transactions occur every second. Detecting fraudulent activities and tracing suspicious funds is a major challenge. Our project, Blockchain Transaction Analysis and Fraud Detection, aims to identify suspicious transactions and accounts by analyzing transaction data using network-based and statistical methods.

The system visualizes how funds move between accounts and detects unusual behavior patterns through a fraud scoring model.

OBJECTIVES

01 To build a directed transaction graph where accounts are nodes and transactions are edges, enabling visualization of fund flows.

02 To detect potentially fraudulent accounts using a composite fraud score based on Znorm (normalized z-score), C (centrality), B (bulk transaction flag), and S (spike flag).

03 To provide an interactive web interface for searching transactions, tracing transaction paths, and visualizing high-risk accounts.

04 To enhance transparency and monitoring by identifying suspicious activity.

METHODOLOGY

Step 1

Data Preparation

- Load transaction CSV files (P1.csv–P5.csv) .
- Normalize key columns: txn_id, uid, from, to, amount.
- Store data per person and combine for analysis.

Step 2

Graph Construction

- A directed graph (DiGraph) is built using NetworkX.

1. Nodes: Represent accounts.
2. Edges: Represent transactions between accounts.
3. Edge attributes: Include transaction IDs, amount, and the person who made the transaction.

- This graph enables visualization of how money flows between accounts and forms the basis for network-based fraud analysis.

Step 3

Transaction Lookup & Path Tracing

- Users can search for a specific transaction ID, retrieving details quickly.
- Users can also trace a transaction path from a source account to a target account using Breadth-First Search (BFS).
- This allows detection of how funds moved through multiple accounts.

Step 4

Fraud Detection

- Each account receives a fraud score based on four metrics:

1. **Znorm (Normalized z-score 0–1):** Measures how unusual a transaction amount is compared to the sender's normal behavior.
2. **C (Centrality):** Measures how influential or connected an account is in the network; highly connected accounts may be higher risk.
3. **B (Bulk transaction flag 0 or 1):** Flags accounts making multiple large-value transactions.
4. **S (Spike flag 0 or 1):** Flags accounts with sudden transaction spikes beyond their usual patterns.

- The metrics are weighted and combined to produce a fraud score from 0–100, highlighting accounts with higher risk.

Step 5

Visualization & Web Interface

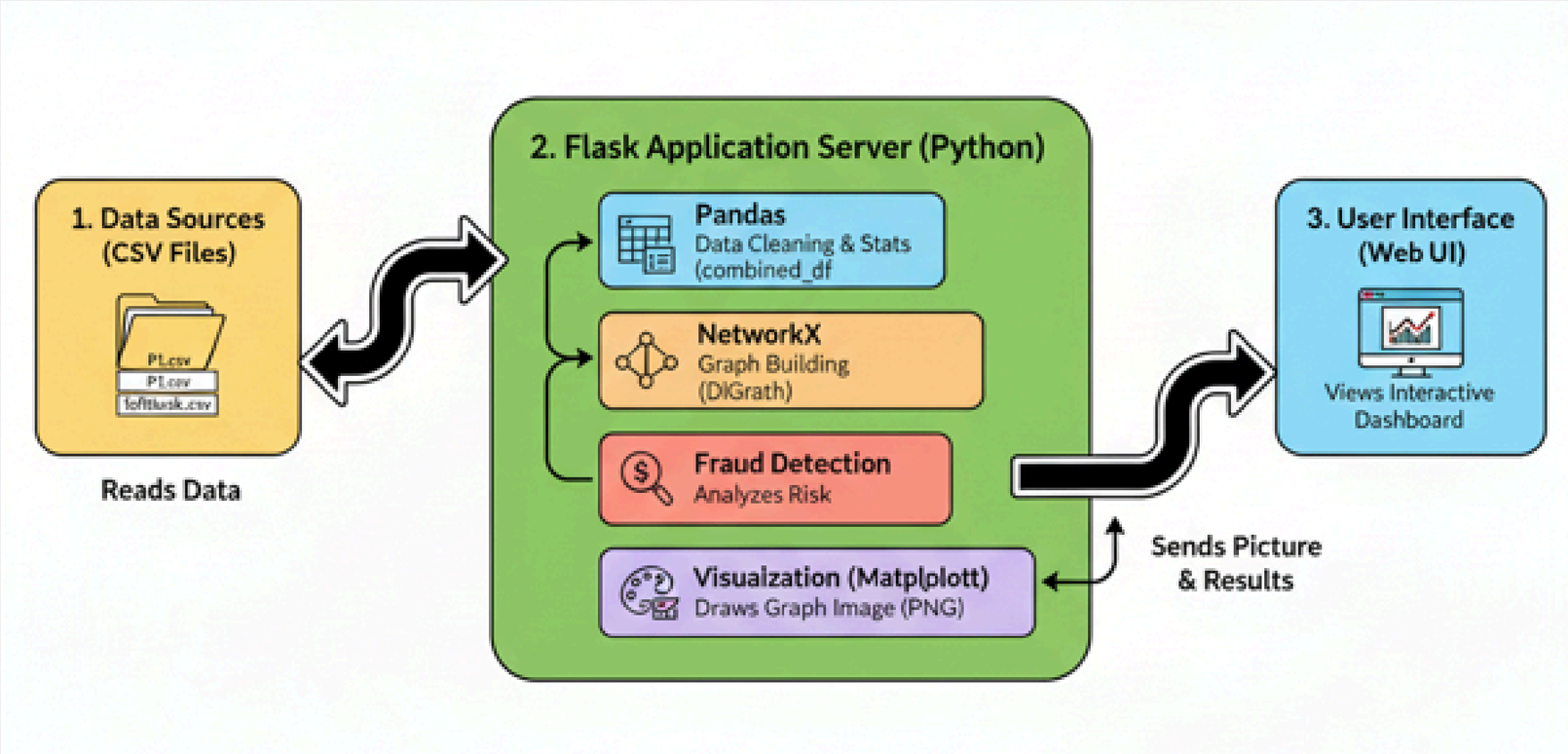
1. Matplotlib is used to visualize the transaction network:

- Nodes: Accounts (colored by risk or selection).
- Edges: Transactions (highlighted if part of a traced path).

2. The Flask web interface provides:

- Graph visualization per person or for all data.
- Transaction search and tracing features.
- Real-time display of fraud scores and high-risk accounts.

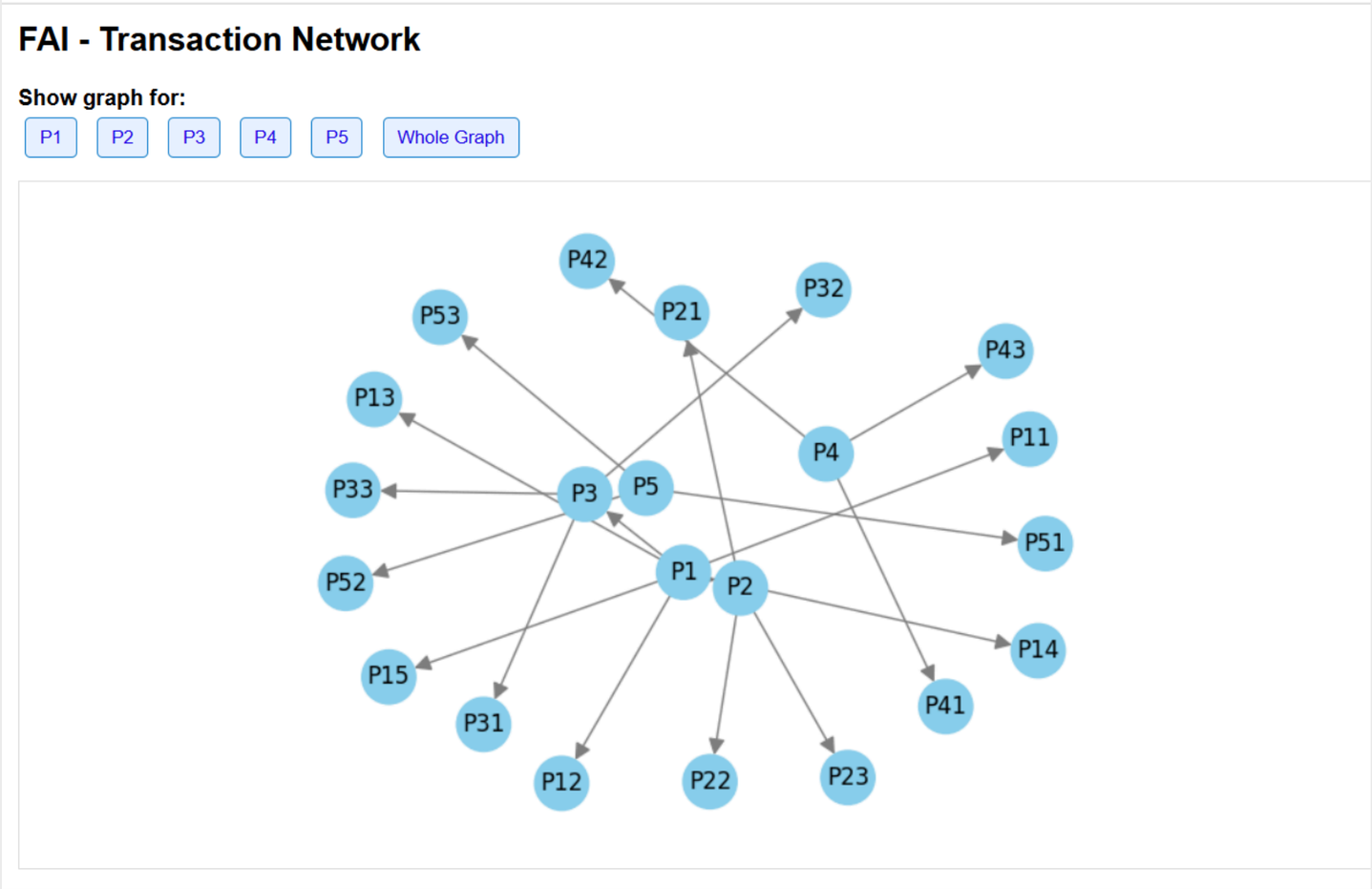
DATA FLOW ARCHITECTURE



Tools and Technologies Used:

Component	Library
Backend Framework	Flask (Python)
Data Handling	Pandas
Network Modeling	NetworkX
Visualization	Matplotlib
Frontend	HTML, JavaScript

RESULT


FAI - Transaction Network
Show graph for:
P1
P2
P3
P4
P5
Whole Graph
P42
P21
P32
P53
P43
P13
P11
P4
P33
P3
P5
P51
P52
P1
P2
P14
P15
P41
P31
P12
P22
P23

CONCLUSION

- The system efficiently analyzes blockchain-style transaction data from multiple users.
- Directed transaction graphs provide a clear visualization of money flows between accounts.
- Fraudulent patterns, such as bulk transactions, spikes, and central accounts, are effectively identified.
- Users can trace paths between accounts and dynamically add new transactions with immediate updates.
- Overall, the project demonstrates a practical approach to fraud detection and transaction monitoring.

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THANK YOU